

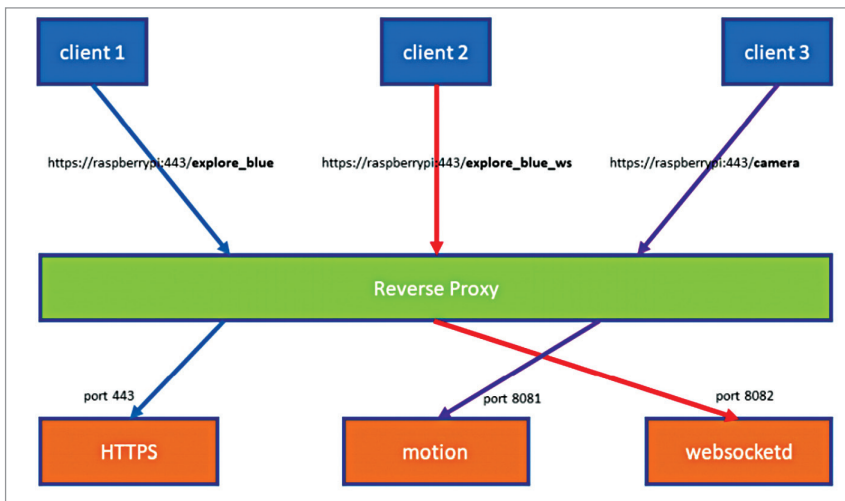


Figure 13: Reverse proxying

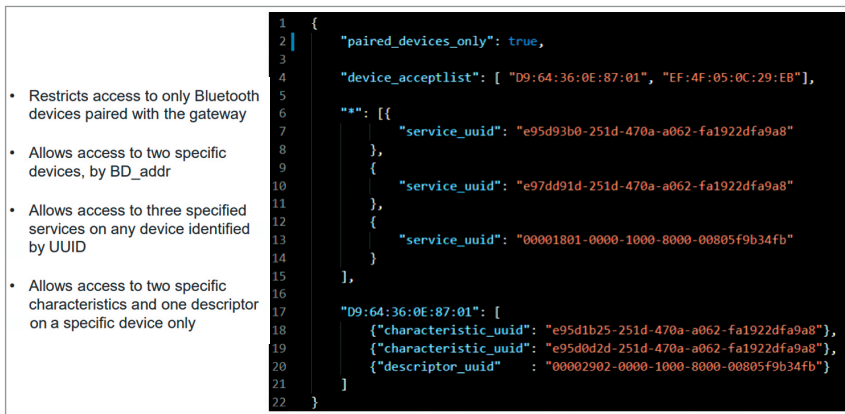


Figure 14: Bluetooth firewall configuration

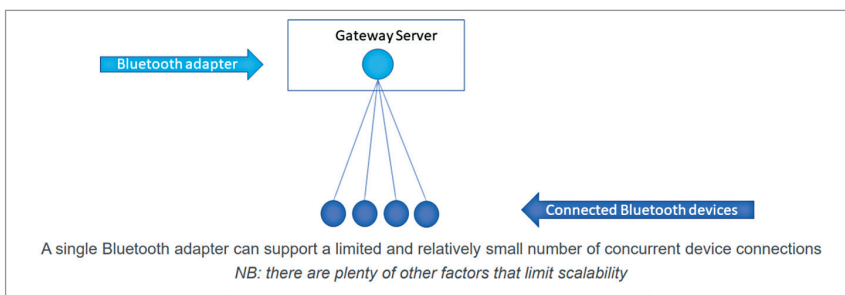


Figure 15: Scalability

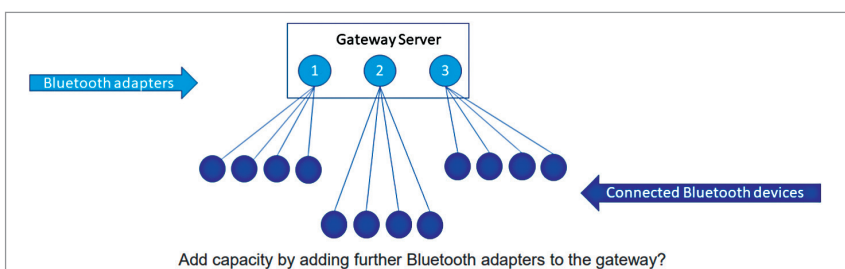


Figure 16: Capacity building with adapters

It was also considered desirable that there be some way to grant access to specific Bluetooth devices only rather than all devices that happened to be in range of the gateway. Further, it would be advantageous to be able to grant access to only specific device capabilities rather than all things that a device can do over Bluetooth.

Reverse proxying

There were three services running on the gateway, each listening to a different port. But the installed IP firewall was configured to only allow port 443 access. The Apache Web Server was set up to act as a reverse proxy and forward port 443 packets to another port directed by the URL patterns (Figure 13). This limited the number of ports exposed to the Internet.

In addition, the concept of a Bluetooth firewall was invented for the project. This allowed control over which Bluetooth devices could be interacted with via the gateway and which of their services, characteristics, and descriptors could be accessed. The configuration file for the Bluetooth firewall is shown in Figure 14.

Bluetooth Internet gateway scalability

Whilst the Bluetooth specifications do not stipulate a limit to the number of concurrent connections that a central device can establish with a peripheral device, there is always a limit in practice, and it can be quite low.

So here are some options if you want to support a large number of devices that are concurrently connected to and controlled by the Bluetooth Internet gateway.

1. Adding multiple adapters: You can add multiple Bluetooth adapters (Figure 16) to your gateway. Some adapters come in the form of USB dongles, so physically that will be easy to implement; some tweaks to your code will be needed but that wouldn't be hard.